MA378 Chapter 5: Review

§5.1 Review (and Exam Preview)

Dr Niall Madden (Niall.Madden@UniversityOfGalway.ie)

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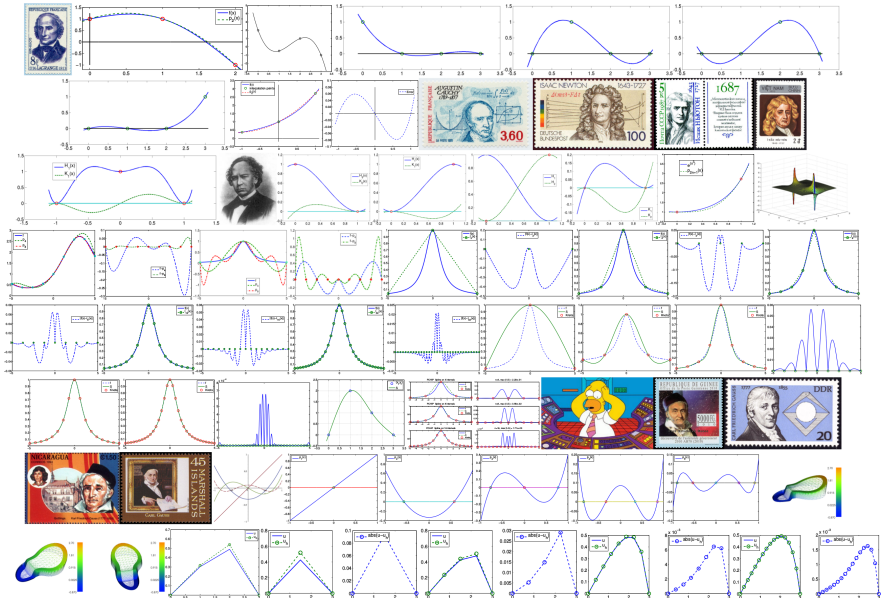

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1.1 Review

This module has four chapters:

1. Polynomial Interpolation
2. Piecewise polynomial interpolation
3. Numerical Integration (Quadrature)
4. Finite element methods

We also studied implementation of methods in MATLAB.

We now review these — primarily to provide you with a check-list of topics.

1.1 The polynomial interpolation problem (PIP) in 2 forms;

1.2 Uniqueness (*if* $p_n \in \mathcal{P}_n$ *has* $n+1$ *zeros, then* $p_n \equiv 0$)

1.2 **Lagrange Polynomials and Lagrange Interpolation**

1.3 Rolle's Theorem

1.3 **Cauchy's theorem**: $f(x) - p_n(x) = \frac{f^{(n+1)}(\tau)}{(n+1)!}\pi_{n+1}(x)$.

1.4 Hermite Interpolation: construction and error estimates.

1.5 Convergence & Runge's Example

2.1 Linear Interpolating Splines:

- construction,
- error analysis,
- best approximation, and
- the Minimum Energy property.

2.2 (Natural) Cubic Splines: construction (error analysis stated without proof)

2.3 PCHIP: Piecewise Cubic Hermite Interpolating Polynomial.

- construction,
- error analysis.

1.1 Review §3 Numerical Integration ("Quadrature")

3.1 Newton-Cotes methods, and the Trapezium Rule

3.1 Undetermined Coefficients

3.2 Simpson's Rule and (non-sharp) error estimates

3.3 Precision

3.3 Composite rules

3.4 Gaussian Quadrature (and *Undetermined Coefficients* again)

3.5 A sequence of Orthogonal Monic Polynomials (and vector spaces and inner products); properties of orthogonal monic polynomials; their roots as quadrature points.

3.6 Precision and convergence of Gaussian Quadrature methods.

1.2 §4: Finite Element Methods

4.1 Boundary value problems, and Maximum Principles.

4.1 The **variational** (weak) **formulation** of the problem, and the uniqueness of its solution.

4.2 The **FEM**, including finite and infinite dimensional spaces.

4.2 **FE implementation**, including Galerkin basis functions ("hat functions"), how to construct the linear system, and why the system matrix is tridiagonal and symmetric.

4.3 Analysis, **Cea's Lemma**.

1.3 Assessment

MA378 is assessed by

- Labs (10%)
- Two Assignments (20%)
- Class test (10%)
- Written two-hour exam (60%)

The exam will have **4 questions**: answer all 4.

Q1 (28 marks) has 4 parts, all concerning theorems and proofs. **Answer any two**. For the Semester 2 2022/2023 exam, the theorems in question are selected from the following:

- **Chapter 1**:
  - 2.3+2.7 (There exists a unique solution to the PIP);
  - 3.3 (Cauchy's Theorem).
- **Chapter 2**:
  - 1.7 (Error is no worse than twice the best possible);
  - 1.9 (Minimum Energy)
- **Chapter 3**:
  - 5.8+5.10 (Monics form a basis, and are orthogonal to all polys of lower degree),
  - 5.15 (Zeros of the monic polys have special properties).
- **Chapter 4**:

- 1.2+1.3 (Max Principle, and unique solution to the BVP),
- 1.6 (Unique solution to the variational problem).

The remaining 3 questions all are worth 24 marks.

Q2: Interpolation (all types)

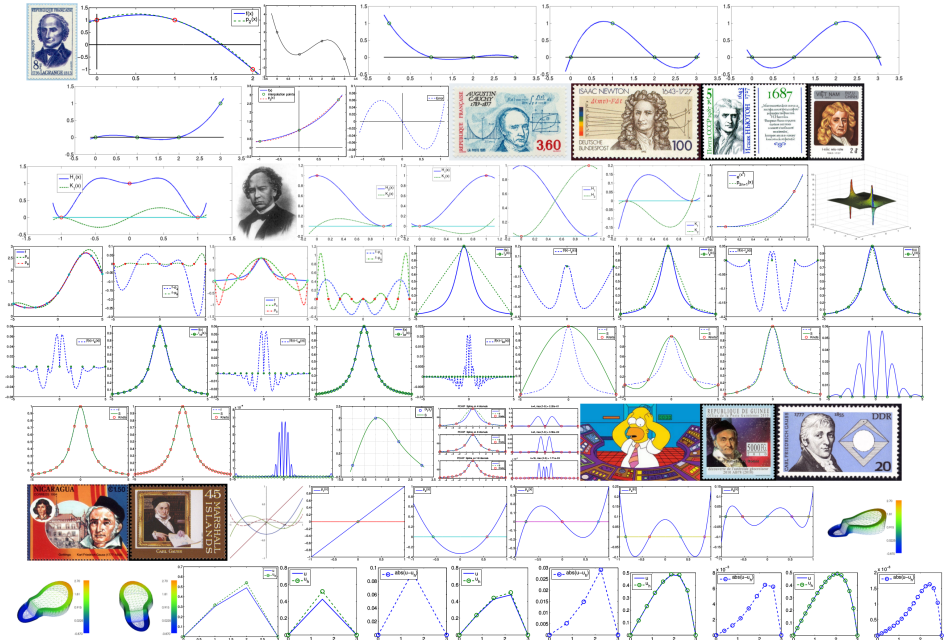
Q3: Quadrature (all types)

Q4: BVPs+FEMs

For each of these, the emphasis is on derivation of methods, and demonstration of understanding, rather than proving theorems (except the occasional uniqueness proof...).

1.4 Thanks

Thanks for taking this module. I hope you have found it interesting, and that some of the topics covered are useful in your future mathematical and computational efforts. Feedback from graduates suggests its one of the most useful "MA" modules. If you are interested in graduate studies in this field, in Ireland or elsewhere, please talk to me.



1.4 Thanks

FYI: The monic polynomials we studied in Chapter 3 are also known as Legendre Polynomials, named after Adrien-Marie Legendre.
This is the only known "portrait" of him, which was (re)discovered in 2008.